

Lab Assignment #1 – Group A

Homework exercises must be completed individually. Solutions must be prepared in the python programming language and present during the lab session on November 25 at 10:30.

1. You are given two sorted sequences which consist of distinct elements. Design an algorithm that finds the k -th smallest element in the union of both sequences. The running time of your algorithm should be $O(\log n + \log m)$ in the worst case where n and m are the sizes of the sequences. To get full credit, you need to provide the efficiency analysis.

2. In a sequence of n integers $a_1, a_2, \dots, a_n$, an inversion is a pair (a_i, a_j) with $i < j$ but $a_i > a_j$. Give an algorithm for computing the number of inversions in a sequence of n integers, that runs in time $O(n \log n)$. To get full credit, you need to provide the efficiency analysis.

You must bring your computers for the presentation. You are expected to explain the solutions you produced for the questions.